

AARYAN MANTRI

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in [linkedin.com/in/aaryan-mantri](https://www.linkedin.com/in/aaryan-mantri)
codeforces.com/profile/mantryan

github.com/DecayDestructor </>

Education

Sardar Patel Institute of Technology

Bachelor of Technology in Computer Engineering

Aug. 2023 – May 2027

CGPA: 9.07

Work Experience

Global Technology Intern

May 2026 – July 2026

Russell Investments

Mumbai, India

- Solo-architected and delivered a **multi-agent AI platform** using LangGraph, LangChain, FastAPI, and React for Russell Investments' US Advisor Intermediary Solutions business, reducing advisor call-preparation time from **30–45 minutes to under 5 minutes**.
- Built a **hybrid RAG pipeline** combining dense vector retrieval and BM25 search across **1,000+ investment documents**, enabling source-grounded responses for product, strategy, and sales-content queries.
- Designed a **query decomposition and routing framework** that classified natural-language requests into structured analytics (Power BI/Snowflake KPI data) and unstructured knowledge retrieval workflows, unifying numerical and semantic QA behind a single interface.
- Developed a **supervisor-agent architecture** coordinating Product, Analytics, and Call-Prep agents, enabling intent-aware routing and automated advisor intelligence generation.

Fullstack Developer Intern

July 2025 – August 2025

SPTBI (Sardar Patel Technology Business Incubator) 

Mumbai, India

- Designed and implemented a role-based **room booking module** for incubatees, supporting both direct paid bookings and admin-approved workflows for restricted rooms; adopted by **110+ startups**.
- Built an internal **ticketing system** enabling incubatees to raise issues or support requests, significantly improving response time and administrative coordination.
- Refactored over **6000 lines** of legacy code, reducing codebase size by **85%** while introducing new features and improving maintainability.

Projects

Distributed Autocomplete | FastAPI, Redis, Kafka, Docker, Nginx, Prometheus, Grafana

April 2026

- Built a **distributed autocomplete service** with a custom **compressed radix trie** maintaining top-k ranked suggestions per node, sharded across **4 Docker containers** using a hybrid **range partitioning + consistent hashing** router
- Engineered a **Kafka streaming pipeline** that decouples read and write paths — completed searches are aggregated over **30-second windows** and asynchronously fed back into trie shards, keeping the query path non-blocking
- Implemented **automatic fault tolerance** via heartbeat health checks; dead shards are removed from the hash ring within **5 seconds** and traffic redistributes to surviving nodes without manual intervention
- Achieved **99% cache hit rate** and **zero failures** across **15,000+ requests** under load testing at **80 req/sec** with 100 concurrent users; observable via live Prometheus + Grafana dashboard

Attendomatic | FastAPI, PostgreSQL, SQLAlchemy, Groq LLM

February 2026

- Built a **platform-independent attendance tracking service** with a decoupled adapter architecture, enabling extension to any messaging platform (Telegram, WhatsApp, Discord) without modifying core business logic
- Engineered an **LLM-powered natural language intent parser** that converts free-text messages into validated JSON actions using structured output schemas, with a confirmation-based execution flow to prevent unintended operations

Technical Skills

Languages: Python, C/C++, JavaScript, HTML/CSS

Technologies: FastAPI, NodeJS, ExpressJS, Docker, Kafka, Nginx, Prometheus, Grafana

Databases: PostgreSQL, MongoDB, Redis

Achievements

Specialist (max. 1421)

Present

Codeforces

Position of Responsibility

Core Team - Training and Placement Office

November 2024 – Present

Statistics Head

Sardar Patel Institute of Technology